# Yongchun Li - Week 7 Report (*1378156*)

## 1. Tasks Accomplished This Week

**Meetings & Collaboration:**

1. Attended daily stand-up meetings (1 h).
2. Participated in the weekly team retro & next week planning(2 h).
3. Participated in Backend Structure and POJO Layering Strategy discussion (20min).

**Development Work: see PR#149**

4. Enhanced logging system implementation with enterprise-grade features (5h):
   - ✓ Implemented comprehensive database persistence for operation logs
   - ✓ Developed asynchronous processing for performance optimization
   - ✓ Created fine-grained control for operation categorization
   - ✓ Enhanced metadata capture for user operations and context
   - ✓ Maintained existing aesthetic console output format
5. Improved enum implementations according to Alibaba Java Coding Guidelines (1.5h):
   - ✓ Created BusinessType, OperatorType, and BusinessStatus enums with proper fields
   - ✓ Implemented helper methods for enum lookup and type safety
   - ✓ Enhanced code documentation and maintainability
6. Reviewed and merged several ready-to-integrate PRs to streamline development flow (10min).

## 2. Key Issues Encountered

The integration of asynchronous log processing required careful handling of thread management to prevent resource leaks. Initial implementation caused occasional task queue backups, which was resolved by implementing proper thread pool configuration and shutdown procedures.

Adding database persistence for logs required significant database schema design considerations, particularly around indexing strategies for optimal query performance while balancing write performance.

The token-based authentication system remains a limitation for the logging system's ability to capture user context fully, which has been documented as a technical limitation requiring future integration.

## 3. Plan for Week 8

Sprint 2 – Remaining Responsibilities:

- Begin implementation of the token-based authentication and authorization system
- Write comprehensive unit tests for the logging system and enhanced enums
- Continue refining system architecture diagram and database design
- Conduct performance testing for the asynchronous logging implementation
- Address code review feedback from team members